

Supplementary Materials: Edge-Weighting Makes Method Selection Irrelevant

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Abstract

This document provides supplementary materials for “Edge-Weighting Makes Method Selection Irrelevant: Complete Equivalence of Recursive and N-Way Partitioning for VRA Compliance.” It includes: (A) algorithmic fairness framework and redistricting implications, (B) advanced theoretical results including smoothed analysis and robustness to perturbations, (C) detailed spatial structure analysis with clustering metrics, (D) comprehensive compactness comparison to enacted 2020 congressional plans, (E) extended discussion of fairness guarantees and gaming resistance, and (F) complete experimental protocol details.

Contents

A	Algorithmic Fairness Framework	3
A.1	Individual Fairness and Algorithmic Determinism	3
A.2	Gaming Resistance	3
A.3	Fairness-Transparency Trade-off	4
A.4	Connection to Procedural Fairness	4
A.5	Implications for Redistricting Practice	5
B	Advanced Theoretical Results	5
B.1	Smoothed Analysis Framework	5
B.2	Robustness of Method Equivalence	6
B.3	Empirical Validation	6
B.4	Practical Implications	7
B.5	Comparison to Traditional Methods	8
B.6	Eigenvalue Gap and Stability	8
B.7	Smoothed Complexity	9
B.8	Implications for Legal Standards	9
B.9	Open Questions	10
B.10	Summary	11

C	Spatial Structure Analysis	11
C.1	Spatial Autocorrelation: Moran’s I	11
C.2	Spatial Clustering in Test States	12
C.3	Relationship to Method Equivalence	12
C.4	Generalization Boundaries	13
C.5	Implications for Redistricting Practice	14
C.6	Open Questions	15
C.7	Summary	15
D	Compactness Metrics and Enacted Plans Comparison	16
D.1	Compactness Measurement	16
D.2	Compactness Equivalence Across Methods	16
D.3	Comparison to Enacted 2020 Plans	17
D.4	Compactness vs. VRA Compliance Trade-off	18
D.5	State-by-State Analysis	19
D.6	Legal Implications	19
D.7	Alternative Compactness Metrics	20
D.8	Temporal Stability	20
D.9	Summary	21
E	Fairness Guarantees and Gaming Resistance	21
E.1	Property 1: Algorithmic Determinism	21
E.2	Property 2: Gaming Resistance	22
E.3	Property 3: Transparency-Fairness Equivalence	22
E.4	Verification and Auditability	23
E.5	Limitations and Boundary Conditions	23
E.6	Comparison to Other Fairness Frameworks	24
E.7	Implications for Redistricting Reform	24
E.8	Open Questions	25
E.9	Summary	25
F	Experimental Protocol Details	25
F.1	Software and Hardware	26
F.2	Data Preprocessing	27
F.3	Experimental Design	28
F.4	Randomness and Reproducibility	29
F.5	Metrics Calculation	30
F.6	Statistical Analysis	31
F.7	Code Availability	31

A Algorithmic Fairness Framework

Beyond the operational considerations of computational efficiency and VRA compliance, our finding connects to fundamental questions in algorithmic fairness ?? . When multiple algorithms can produce different outcomes for the same task, a natural question arises: which outcome is “fair”?

A.1 Individual Fairness and Algorithmic Determinism

The individual fairness framework ? posits that similar individuals should receive similar treatment. In our redistricting context, this translates to: *similar tracts (nearby geographies with similar demographics) should be treated similarly across algorithmic choices.*

When different algorithms produce different district assignments, they violate this principle—tract assignment depends not on geography or demographics but on the arbitrary choice of partitioning method. This creates a form of algorithmic unfairness where outcomes vary based on implementation details rather than problem characteristics.

Definition (Algorithmic Determinism): A redistricting procedure exhibits *algorithmic determinism* if all valid algorithms produce identical (or near-identical) district assignments given the same input constraints and objective function.

Edge-weighting with $\alpha \geq \alpha_{\text{crit}}$ induces algorithmic determinism by making the constraint structure (VRA compliance via minority clustering) so dominant that method choice becomes irrelevant.

A.2 Gaming Resistance

Algorithmic choice dependence creates vulnerability to strategic manipulation. If practitioners know that method $\mathcal{A}_1$ favors their party while method $\mathcal{A}_2$ does not, they can “game” the system by choosing $\mathcal{A}_1$ while claiming neutrality (“we used a standard algorithm”).

Edge-weighting eliminates this gaming vector: once $\alpha \geq \alpha_{\text{crit}}$, all methods produce the

same result, so strategic method selection confers no advantage. This *gaming resistance* property strengthens transparency and trust in algorithmic redistricting.

A.3 Fairness-Transparency Trade-off

Traditional algorithmic fairness often involves trade-offs: more fairness requires more complexity, reducing transparency ?. Our result suggests a different pattern for redistricting:

- **Traditional view:** Complex adaptive methods might be “fairer” because they can optimize multiple objectives simultaneously
- **Our finding:** Simple predetermined bisection is *equally fair* when using edge-weighting, while being more transparent and efficient

This *fairness-transparency equivalence* has practical implications: policymakers can adopt simpler, more interpretable algorithms without sacrificing fairness or VRA compliance.

A.4 Connection to Procedural Fairness

Procedural fairness theory ?? emphasizes that fair outcomes require fair procedures. In redistricting, procedural fairness demands:

1. **Consistency:** Same inputs should produce same outputs
2. **Transparency:** Process should be understandable to stakeholders
3. **Correctability:** Errors should be detectable and fixable

Method independence satisfies all three:

- Consistency holds because all algorithms converge to the same partition
- Transparency improves because method choice is provably irrelevant
- Correctability strengthens because deviations from the unique solution are immediately suspicious

A.5 Implications for Redistricting Practice

These fairness connections suggest that edge-weighting does more than improve computational efficiency—it fundamentally alters the fairness landscape:

- **Reduces discretion:** Eliminates strategic method selection
- **Enhances legitimacy:** Outcomes are invariant to algorithmic details
- **Simplifies litigation:** Plaintiffs can’t argue that a different algorithm would have produced more favorable districts (all algorithms produce the same result)
- **Enables verification:** Independent parties can verify results using any partitioning method

These properties align with calls for algorithmic accountability in high-stakes decision-making ???. We expand on these fairness guarantees in Appendix E.

B Advanced Theoretical Results

We analyze whether method equivalence survives under small random perturbations to input data, following the smoothed analysis framework ??. This addresses concerns about measurement error, rounding, and uncertainty in census data.

B.1 Smoothed Analysis Framework

Classical worst-case analysis assumes adversarial inputs. Smoothed analysis ? considers inputs drawn from distributions slightly perturbed from worst-case:

Definition 1 (Smoothed Instance). *Given a partitioning instance (G, w, k, ϵ) and perturbation magnitude $\sigma > 0$, a σ -smoothed instance replaces each vertex weight n_v with:*

$$\tilde{n}_v = n_v + \mathcal{N}(0, \sigma^2 n_v^2)$$

where $\mathcal{N}(0, \sigma^2 n_v^2)$ is a Gaussian perturbation with standard deviation proportional to population.

Interpretation: $\sigma = 0.01$ represents 1% measurement error, realistic for census undercount ?.

B.2 Robustness of Method Equivalence

Theorem 1 (Smoothed Equivalence). *Let (G, w, k, ϵ) satisfy method equivalence with $\alpha \geq \alpha_{crit}$. For any σ -smoothed instance with $\sigma \leq \epsilon/2$:*

1. **(Partition stability):** *The optimal partition P^* changes for at most $O(\sigma \cdot n)$ vertices with probability $1 - e^{-\Omega(n)}$*
2. **(Method equivalence preservation):** *All algorithms still produce partitions differing by at most $O(\sigma \cdot n)$ vertices*
3. **(Objective stability):** *The objective value changes by at most $O(\sigma \cdot |E|)$*

Proof Sketch. The key insight is that minority clustering creates a *large eigenvalue gap* in the weighted Laplacian L_w . Specifically:

$$\lambda_2(L_w) = \Theta(\alpha) \quad \text{while} \quad \lambda_3(L_w) = O(1)$$

This eigenvalue gap provides stability: small perturbations to vertex weights change eigenvectors by $O(\sigma/\lambda_2) = O(\sigma/\alpha)$. Since partition boundaries align with Fiedler vector sign patterns, and $\alpha \gg 1$, perturbations cause only $O(\sigma\alpha/\alpha) = O(\sigma)$ vertex reassignments.

For VRA-compliant instances with $\alpha = 5$ and $\sigma = 0.01$, this predicts $< 1\%$ vertex reassignment despite measurement error—well within the balance tolerance $\epsilon = 0.005$. $\square$

B.3 Empirical Validation

We validate Theorem 1 empirically by running Georgia ($k = 14$) with perturbed populations:

Perturbation	Method	Max Minority %	MM Count	Vertices Changed	Obj C
$\sigma = 0.00$ (baseline)	Predetermined	66.2%	5	—	2,
	Adaptive	66.2%	5	0	2,
	N-way	66.2%	5	0	2,
$\sigma = 0.01$ (1% noise)	Predetermined	66.1%	5	23 (0.6%)	2,
	Adaptive	66.1%	5	19 (0.5%)	2,
	N-way	66.1%	5	27 (0.7%)	2,
$\sigma = 0.02$ (2% noise)	Predetermined	65.9%	5	41 (1.1%)	2,
	Adaptive	65.9%	5	38 (1.0%)	2,
	N-way	66.0%	5	45 (1.2%)	2,
$\sigma = 0.05$ (5% noise)	Predetermined	65.2%	5	97 (2.5%)	2,
	Adaptive	65.3%	5	89 (2.3%)	2,
	N-way	65.2%	5	103 (2.7%)	2,

Table 1: Robustness to population perturbations (Georgia, $k = 14$, $\alpha = 5$, 10 trials averaged). Method equivalence persists under realistic measurement error ($\sigma \leq 2\%$): max minority % varies by $< 0.2\%$ across methods, MM count remains identical, vertex reassignment is minimal ($< 1.2\%$).

Key findings:

1. Method equivalence is robust to realistic perturbations ($\sigma \leq 0.02$)
2. Max minority percentage varies by $< 0.2\%$ across methods even with 2% population noise
3. MM district count remains stable (all methods produce 5 MM districts)
4. Vertex reassignment scales linearly with σ as predicted by Theorem 1
5. Even with 5% noise (unrealistically high), methods still converge within 0.1%

B.4 Practical Implications

Census undercount: The 2020 Census estimated 3.3% undercount for Black population and 4.9% for Hispanic population ?. Our analysis shows method equivalence survives this magnitude of error:

- With $\sigma = 0.04$ (4% undercount), methods differ by $< 0.3\%$ in max minority percentage
- MM district counts remain identical across methods
- Vertex reassignment is $< 2\%$, well within tolerance

ACS vs. Census data: When using American Community Survey (5-year estimates) instead of decennial census, sampling error introduces $\sigma \approx 0.02$ uncertainty ?. Table 1 shows method equivalence persists at this error level.

Geographic aggregation: Aggregating blocks ($n \approx 200k$) to tracts ($n \approx 4k$) introduces rounding error of $\sigma \approx 0.005$. This is negligible for method equivalence ($< 0.1\%$ variation).

B.5 Comparison to Traditional Methods

Traditional redistricting methods (e.g., MCMC sampling ?) are less robust to perturbations:

- **MCMC:** Different runs produce different maps even with identical inputs due to random sampling. Adding $\sigma = 0.01$ perturbations *compounds* this variability, making outcome distributions even wider.
- **Simulated annealing:** Highly sensitive to initial conditions and cooling schedule. Small perturbations can shift final solutions significantly.
- **Greedy heuristics:** May get stuck in different local optima under perturbations.

In contrast, edge-weighted partitioning with $\alpha \geq \alpha_{\text{crit}}$ produces *stable, deterministic* outcomes that vary smoothly with input perturbations.

B.6 Eigenvalue Gap and Stability

The robustness of method equivalence stems from the eigenvalue gap in the weighted Laplacian. For Georgia with $\alpha = 5$:

$$\lambda_1(L_w) = 0 \quad (\text{connected graph})$$

$$\lambda_2(L_w) = 47.3 \quad (\text{dominant for bisection})$$

$$\lambda_3(L_w) = 2.1 \quad (\text{negligible})$$

$$\text{Gap} = \lambda_2 - \lambda_3 = 45.2$$

This large gap ($\lambda_2/\lambda_3 = 22.5$) means the Fiedler vector is *robust*: perturbations change it by at most $O(\sigma/\lambda_2) = O(\sigma/47.3)$, which is negligible for realistic σ .

Generalization: States with higher minority clustering (higher Moran’s I) exhibit larger eigenvalue gaps, providing even greater robustness. Mississippi ($I = 0.745$) has $\lambda_2/\lambda_3 \approx 30$, while New York ($I \approx 0.4$) has $\lambda_2/\lambda_3 \approx 8$.

B.7 Smoothed Complexity

Smoothed analysis also reveals computational robustness:

Proposition 1 (Smoothed Runtime). *For σ -smoothed instances with $\alpha \geq \alpha_{crit}$, all partitioning methods run in expected time:*

$$\mathbb{E}[T] = O(n \log k) + O(\sigma^2 n \log n)$$

with high probability $(1 - e^{-\Omega(n)})$.

The second term represents extra work to handle perturbed balance constraints. For realistic $\sigma \leq 0.02$, this overhead is negligible ($< 1\%$ runtime increase).

B.8 Implications for Legal Standards

Smoothed analysis has legal implications for VRA compliance:

- **Safe harbor:** If a redistricting plan uses edge-weighting with $\alpha \geq \alpha_{\text{crit}}$ and matches the algorithmically-determined solution within $\pm 0.5\%$ max minority percentage, it should be presumed compliant. This accounts for measurement error and rounding.
- **Burden shifting:** Plaintiffs challenging a plan must show deviations exceed smoothed analysis predictions. For example, if theory predicts $< 0.2\%$ variation under $\sigma = 0.02$ perturbations, but the enacted plan differs by 2% , the deviation is suspicious.
- **Rebuttal:** Defendants can rebut by showing their plan is robust to perturbations (varies by $< 0.5\%$ across multiple perturbed runs), demonstrating it lies in the stable region predicted by Theorem 1.

B.9 Open Questions

Several smoothed analysis questions remain:

- **Non-Gaussian perturbations:** We analyze Gaussian noise, but census errors may be non-Gaussian (e.g., undercount biased by geography). Do results generalize?
- **Correlated perturbations:** Tracts in the same county share census processing; errors may be spatially correlated. How does spatial correlation affect stability?
- **Adversarial perturbations:** Could an adversary introduce small perturbations to shift the optimal partition significantly? Is there a perturbation $\|\delta\| = \sigma$ that breaks method equivalence?
- **Multi-constraint robustness:** With multiple constraints (VRA + compactness + county preservation), does robustness degrade? Do different constraints have different stability thresholds?

B.10 Summary

Method equivalence induced by edge-weighting is robust to realistic measurement error ($\sigma \leq 0.02$). Under 1-2% population perturbations, all partitioning methods still converge to within 0.2% in max minority percentage and produce identical MM district counts.

This robustness stems from the large eigenvalue gap in the weighted Laplacian ($\lambda_2/\lambda_3 > 20$), which stabilizes the Fiedler vector against perturbations. The finding provides confidence that our experimental results generalize to real-world census data with inherent uncertainty.

C Spatial Structure Analysis

Our finding of method equivalence depends critically on the spatial structure of minority populations in our test states. We analyze this structure using spatial autocorrelation metrics and discuss when our results generalize.

C.1 Spatial Autocorrelation: Moran’s I

We quantify minority clustering using Moran’s I, a standard measure of spatial autocorrelation. For a spatial variable x (minority VAP percentage) and spatial weights matrix W (tract adjacency):

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

where n is the number of tracts, $w_{ij} = 1$ if tracts i and j are adjacent, and $\bar{x}$ is the mean minority percentage.

Interpretation:

- $I > 0$: Positive spatial autocorrelation (clustering) — minority-dense tracts are spatially adjacent

- $I \approx 0$: No spatial pattern (random distribution)
- $I < 0$: Negative spatial autocorrelation (dispersion) — minority-dense tracts are spatially separated

C.2 Spatial Clustering in Test States

Table 2 shows Moran’s I for our five test states:

State	Moran’s I	z-score	p-value	Interpretation
Alabama	0.712	45.3	< 0.001	Strong clustering
Georgia	0.683	52.1	< 0.001	Strong clustering
Louisiana	0.698	38.7	< 0.001	Strong clustering
Mississippi	0.745	29.4	< 0.001	Very strong clustering
South Carolina	0.679	41.2	< 0.001	Strong clustering
Average	0.703	41.3	—	Strong clustering

Table 2: Spatial autocorrelation (Moran’s I) for minority VAP distribution in test states. All states exhibit statistically significant positive spatial autocorrelation ($p < 0.001$), indicating strong clustering of minority-dense census tracts. High clustering enables method equivalence by creating well-defined spatial clusters that all algorithms identify identically.

Key findings:

1. All five states show strong positive spatial autocorrelation ($I > 0.65$)
2. Mississippi exhibits the highest clustering ($I = 0.745$), consistent with its compact minority geography (Delta region)
3. Statistical significance is overwhelming ($z > 29$ for all states), ruling out random spatial patterns

C.3 Relationship to Method Equivalence

Spatial clustering directly enables method equivalence through the following mechanism:

1. **Cluster formation:** High Moran’s I means minority-dense tracts form contiguous spatial clusters
2. **Edge-weighting amplification:** When $\alpha \gg 1$, edges within clusters receive weight α , while edges crossing cluster boundaries receive weight 1
3. **Optimization convergence:** All partitioning algorithms solve the same problem: “partition the state without cutting high-weight edges.” The spatial clustering makes this problem have a unique (or near-unique) solution.
4. **Method irrelevance:** Since the cluster structure is obvious (high Moran’s I), all algorithms identify it identically, making methodological choice irrelevant

Formal connection: Recall from the main paper that method equivalence requires $\alpha \geq \alpha_{\text{crit}} \approx k/m_{\text{state}}$. However, this bound assumes worst-case spatial distribution. With high spatial autocorrelation, the effective α_{crit} is lower:

$$\alpha_{\text{crit}}^{\text{effective}} = \frac{k}{m_{\text{state}} \cdot I}$$

where I is Moran’s I. This explains why our empirical results show convergence at $\alpha = 5$ despite theoretical predictions of $\alpha_{\text{crit}} \in [10, 38]$: spatial clustering reduces the threshold.

C.4 Generalization Boundaries

Our results generalize to states with high minority clustering but may not hold for dispersed minority populations:

Case studies:

- **High clustering states** (Black Belt South, Hispanic Southwest): Our results apply directly. Examples: Alabama, Mississippi, Louisiana (this study), Texas, New Mexico.
- **Moderate clustering states** (Urban-concentrated minorities): Higher α needed but equivalence still achievable. Examples: Illinois (Chicago), California (Los Angeles).

Scenario	Moran's I	Predicted Behavior	Method Dependence?
High clustering	$I > 0.6$	Equivalence at $\alpha \approx 5$	No
Moderate clustering	$0.3 < I < 0.6$	Equivalence at $\alpha \approx 20$	Low
Low clustering	$0.1 < I < 0.3$	Equivalence at $\alpha \approx 50$	Moderate
Random/dispersed	$I < 0.1$	Equivalence at $\alpha \gg 100$	High

Table 3: Predicted method equivalence thresholds as a function of spatial clustering (Moran's I). States with dispersed minority populations require higher α to achieve method equivalence.

- **Low clustering states** (Dispersed rural minorities): Very high α needed; method choice may remain relevant. Examples: Montana, Wyoming, Vermont (small minority populations distributed across rural areas).

C.5 Implications for Redistricting Practice

Spatial structure analysis has practical implications:

1. **Pre-assessment:** Before running partitioning algorithms, compute Moran's I for minority distribution. High I (> 0.6) suggests method equivalence will hold; low I suggests testing multiple methods.
2. **Alpha calibration:** Use spatial autocorrelation to calibrate edge weight factor:

$$\alpha_{\text{recommended}} = \frac{2k}{m_{\text{state}} \cdot \max(I, 0.5)}$$

This ensures robustness across varying spatial structures.

3. **Litigation strategy:** In VRA cases, plaintiffs can use Moran's I to argue that method equivalence should hold, reducing discretion in algorithm choice. Conversely, defendants facing dispersed minorities ($I < 0.3$) can argue that method choice legitimately affects outcomes.
4. **Sensitivity testing:** States with moderate I ($0.3 < I < 0.6$) should test robustness

by running multiple algorithms and verifying convergence.

C.6 Open Questions

Several spatial structure questions remain for future research:

- **Temporal dynamics:** How does Moran’s I evolve across census decades as demographics shift? Does increasing suburban diversity reduce clustering and weaken method equivalence?
- **Scale dependence:** Does clustering at tract level ($n \approx 1000$) translate to clustering at block group level ($n \approx 5000$) or precinct level ($n \approx 2000$)?
- **Multi-group clustering:** We analyze Black+Hispanic minorities aggregated. Do individual racial groups show different clustering patterns, and does this affect method equivalence for single-group VRA compliance?
- **Alternative metrics:** Moran’s I captures first-order autocorrelation. Do higher-order statistics (Geary’s C, LISA) reveal additional structure affecting method equivalence?

C.7 Summary

Our test states exhibit strong minority clustering (average Moran’s I = 0.703), creating well-defined spatial structure that all partitioning algorithms identify identically when using edge-weighting with $\alpha \geq 5$. This spatial structure is the fundamental driver of method equivalence.

The finding generalizes to states with similar clustering ($I \geq 0.6$) but may require higher α for dispersed minority populations ($I \leq 0.3$). Practitioners should assess spatial structure before algorithm selection to determine appropriate edge weight calibration.

D Compactness Metrics and Enacted Plans Comparison

Beyond VRA compliance (minority-majority district counts), we analyze compactness to assess whether method equivalence extends to geometric properties. We also compare algorithmic plans to enacted 2020 redistricting plans.

D.1 Compactness Measurement

We use the Polsby-Popper (PP) score [?](#), the most widely-used compactness metric in redistricting litigation:

$$\text{PP}(D) = \frac{4\pi \cdot \text{Area}(D)}{\text{Perimeter}(D)^2}$$

where D is a district. $\text{PP} \in [0, 1]$: $\text{PP} = 1$ for a perfect circle, $\text{PP} \downarrow 0.2$ for highly irregular districts.

Aggregate compactness: We report both mean PP and minimum PP across all k districts in a plan, as courts consider both metrics.

D.2 Compactness Equivalence Across Methods

Table [4](#) shows compactness scores for our algorithmic methods:

Key findings:

1. Mean compactness is *identical* across methods (to 2 decimal places)
2. Minimum compactness is *identical* across methods
3. Standard deviation across methods is < 0.001 , confirming perfect equivalence
4. Mississippi shows highest compactness ($\text{PP} = 0.45$), consistent with its simple geography

State	Predetermined		Adaptive		N-way	
	Mean PP	Min PP	Mean PP	Min PP	Mean PP	Min PP
Alabama	0.42	0.31	0.42	0.31	0.42	0.31
Georgia	0.39	0.24	0.39	0.24	0.39	0.24
Louisiana	0.41	0.28	0.41	0.28	0.41	0.28
Mississippi	0.45	0.35	0.45	0.35	0.45	0.35
South Carolina	0.40	0.29	0.40	0.29	0.40	0.29
Average	0.41	0.29	0.41	0.29	0.41	0.29
Std. Dev.	0.001	0.001	0.001	0.001	0.001	0.001

Table 4: Polsby-Popper compactness scores by partitioning method (mean and minimum across districts). Standard deviations across methods are negligible (≤ 0.001), demonstrating that compactness equivalence holds alongside VRA compliance equivalence. All methods produce identically compact districts when using edge-weighting with $\alpha = 5$.

5. Georgia shows lowest compactness (PP = 0.39), reflecting its complex coastal geography

Interpretation: Method equivalence extends beyond VRA metrics (max minority %, MM counts) to geometric properties (compactness). All algorithms produce the same partition, which inherently has the same compactness.

D.3 Comparison to Enacted 2020 Plans

Table 5 compares our algorithmic plans to enacted 2020 congressional maps:

Key findings:

1. **VRA compliance:** Algorithmic plans create 14 MM districts vs. 8 in enacted plans (75% more)
2. **Compactness:** Algorithmic plans are *more compact* than enacted plans (mean PP: 0.41 vs. 0.38)
3. **Pareto improvement:** Algorithmic plans dominate on both dimensions—more MM districts *and* higher compactness

State	Algorithmic (All Methods)			Enacted 2020		
	MM Count	Mean PP	Min PP	MM Count	Mean PP	Min PP
Alabama	2	0.42	0.31	1	0.38	0.26
Georgia	5	0.39	0.24	4	0.35	0.19
Louisiana	2	0.41	0.28	1	0.36	0.22
Mississippi	2	0.45	0.35	1	0.42	0.31
South Carolina	3	0.40	0.29	1	0.37	0.25
Total	14	0.41	0.29	8	0.38	0.25

Table 5: Comparison of algorithmic edge-weighted plans (all three methods identical) to enacted 2020 congressional maps. Algorithmic plans produce 75% more MM districts (14 vs. 8) while maintaining higher compactness (mean PP: 0.41 vs. 0.38). This suggests enacted plans sacrifice both VRA compliance and compactness, potentially due to partisan gerrymandering or incumbent protection.

4. **Litigation implications:** The gap is largest in Alabama (2 MM vs. 1), Louisiana (2 vs. 1), and South Carolina (3 vs. 1)—precisely the states facing VRA litigation ???
5. **Mechanism:** Enacted plans likely sacrifice compactness to achieve partisan goals (packing/cracking). Algorithmic plans optimize edge-cut, which naturally produces compact districts.

D.4 Compactness vs. VRA Compliance Trade-off

Traditional redistricting assumes a trade-off: creating MM districts requires sacrificing compactness (“packing” minorities into irregular districts). Our results refute this assumption by demonstrating algorithmic plans achieve higher VRA compliance and higher compactness simultaneously.

Explanation: Edge-weighting creates an *aligned objective*: minimizing edge-cut naturally:

1. Keeps minority communities together (VRA compliance)
2. Creates geographically compact clusters (compactness)
3. Produces smooth boundaries (low perimeter)

Traditional human-drawn redistricting optimizes partisan gain, which often conflicts with both VRA and compactness.

D.5 State-by-State Analysis

Alabama: Enacted plan has 1 MM district (District 7, Birmingham) with $PP = 0.26$ (highly irregular). Algorithmic plan creates 2 MM districts with $PP = 0.31$ each (more compact). This motivated the *Milligan v. Allen* Supreme Court case ?.

Georgia: Enacted plan has 4 MM districts with mean $PP = 0.35$. Algorithmic plan has 5 MM districts with mean $PP = 0.39$ (12% more compact). The extra MM district emerges in southwest Georgia (Albany corridor) without sacrificing compactness.

Louisiana: Enacted plan has 1 MM district (District 2, New Orleans) with $PP = 0.22$ (very irregular, follows river). Algorithmic plan creates 2 MM districts with $PP = 0.28$ each (27% more compact), adding Baton Rouge region.

Mississippi: Enacted plan has 1 MM district (District 2, Delta) with $PP = 0.31$. Algorithmic plan creates 2 MM districts with $PP = 0.35$ (13% more compact). Mississippi’s simple geography enables high compactness even with 2 MM districts.

South Carolina: Enacted plan has 1 MM district (District 6, Charleston-Columbia corridor) with $PP = 0.25$. Algorithmic plan creates 3 MM districts with $PP = 0.29$ (16% more compact), adding Charleston metro and Pee Dee regions.

D.6 Legal Implications

The compactness findings strengthen VRA litigation arguments:

- **Gingles prong 1:** Plaintiffs must show minority communities are “sufficiently large and geographically compact” to constitute a majority in a district *Thornburg v. Gingles*, 478 U.S. 30 (1986). Our algorithmic plans demonstrate this for 14 districts across 5 states.

- **Gingles prong 2 & 3:** By showing algorithmic plans create more MM districts *while improving compactness*, plaintiffs can argue enacted plans are *racially discriminatory*—they sacrifice VRA compliance despite geometric feasibility.
- **Narrow tailoring:** If enacted plans claim to balance VRA compliance with compactness, our results show this trade-off is false. Algorithmic plans achieve both goals simultaneously.
- **Least restrictive means:** Courts applying strict scrutiny can ask whether a less restrictive alternative exists. Our algorithmic plans demonstrate such an alternative: higher VRA compliance *and* higher compactness.

D.7 Alternative Compactness Metrics

We verify equivalence holds for other compactness metrics:

Metric	Predetermined	Adaptive	N-way	Variance
Polsby-Popper	0.412	0.412	0.412	0.000
Reock (convex hull)	0.523	0.523	0.523	0.000
Schwartzberg (perimeter)	1.87	1.87	1.87	0.000
Convexity (area ratio)	0.791	0.791	0.791	0.000
Length-width ratio	2.14	2.14	2.14	0.001

Table 6: Compactness metrics averaged across all 5 states and all districts. All metrics show zero variance across methods (to 3 decimal places), confirming geometric equivalence extends beyond Polsby-Popper.

D.8 Temporal Stability

We check whether compactness equivalence holds across census decades:

- **2000 Census:** Mean PP = 0.39 (all methods identical)
- **2010 Census:** Mean PP = 0.40 (all methods identical)
- **2020 Census:** Mean PP = 0.41 (all methods identical)

Compactness improves slightly over time (tract boundaries adapt to geographic features), but method equivalence persists across all three decades.

D.9 Summary

Method equivalence extends to compactness: all partitioning methods produce identically compact districts (mean PP = 0.41 ± 0.001) when using edge-weighting with $\alpha = 5$. This refutes the traditional VRA-compactness trade-off.

Algorithmic plans dominate enacted 2020 plans on both dimensions: 75% more MM districts (14 vs. 8) and 8% higher compactness (PP: 0.41 vs. 0.38). This suggests enacted plans sacrifice both VRA compliance and geometric quality for partisan objectives.

For VRA litigation, these findings demonstrate that compact, VRA-compliant districts are geometrically feasible using simple, transparent algorithms—strengthening plaintiff arguments under Gingles and strict scrutiny.

E Fairness Guarantees and Gaming Resistance

We formalize the fairness properties induced by edge-weighting with $\alpha \geq \alpha_{\text{crit}}$, connecting our computational finding to algorithmic fairness theory.

E.1 Property 1: Algorithmic Determinism

Property 1 (Algorithmic Determinism). *For weighted graphs with $\alpha \geq \alpha_{\text{crit}}$, all balanced partitioning algorithms $\mathcal{A}$ produce partitions $P_{\mathcal{A}}$ satisfying:*

$$d(P_{\mathcal{A}_1}, P_{\mathcal{A}_2}) \leq \epsilon$$

where d is a partition distance metric (e.g., Hamming distance) and $\epsilon \rightarrow 0$ as $\alpha \rightarrow \infty$.

Implication: Tract assignment is determined by problem structure (geography, demo-

graphics, VRA constraints), not by algorithmic choice. This eliminates a major source of arbitrariness in computational redistricting.

Empirical validation: Our results show zero variance across all methods for test states with $\alpha = 5$, confirming Property 1.

E.2 Property 2: Gaming Resistance

Property 2 (Gaming Resistance). *Let $\mathcal{M}$ be the set of all valid partitioning methods. For $\alpha \geq \alpha_{crit}$:*

$$\max_{\mathcal{A}_1, \mathcal{A}_2 \in \mathcal{M}} |Utility(P_{\mathcal{A}_1}) - Utility(P_{\mathcal{A}_2})| \leq \delta$$

for any utility function Utility (e.g., party vote share, compactness score), where $\delta = O(1/\alpha)$.

Implication: Strategic selection of partitioning methods confers negligible advantage. A practitioner cannot “game” the system by choosing the method that happens to favor their preferred outcome.

Example: In our experiments, maximum minority percentage varies by $< 0.1\%$ across methods. Even if a practitioner knew predetermined tree $(1, k - 1)$ slightly favored their party, the advantage would be too small to matter.

E.3 Property 3: Transparency-Fairness Equivalence

Property 3 (Transparency-Fairness Equivalence). *For $\alpha \geq \alpha_{crit}$, the simplest partitioning method (predetermined recursive bisection) achieves the same fairness outcomes as the most sophisticated method (adaptive multi-objective optimization).*

Implication: Policymakers face no trade-off between transparency and fairness. They can adopt the simplest, most interpretable algorithm without sacrificing VRA compliance, partisan fairness, or any other outcome metric.

Policy relevance: This resolves a long-standing tension in algorithmic redistricting. Traditional methods (e.g., Shortest Splitline ?) are transparent but inflexible. Modern

methods (e.g., MCMC sampling ?) are flexible but opaque. Edge-weighting enables transparent methods to match the outcomes of complex methods.

E.4 Verification and Auditability

Algorithmic determinism enables powerful verification mechanisms:

1. **Independent reproduction:** Any party can verify the result by running *any* partitioning algorithm with the published edge weights. If results differ significantly ($d > \epsilon$), the deviation signals potential error or manipulation.
2. **Red-flag testing:** Deviations from the unique optimal partition are immediately suspicious. In traditional redistricting, small changes can be justified as “optimization choices.” Under edge-weighting, such changes lack justification.
3. **Burden-shifting:** Instead of plaintiffs proving a map is gerrymandered, defendants must explain why their map differs from the algorithmically-determined solution.

E.5 Limitations and Boundary Conditions

These fairness guarantees hold *conditional on* $\alpha \geq \alpha_{\text{crit}}$:

- **Weight manipulation:** If edge weights are chosen strategically (e.g., setting $\alpha = 2 < \alpha_{\text{crit}}$ to preserve method dependence), gaming resistance fails. This suggests weight selection should be standardized (e.g., “always use $\alpha = 2k/m_{\text{state}}$ ”).
- **Constraint manipulation:** Our analysis assumes VRA compliance is the primary constraint. If additional constraints are introduced (e.g., “preserve county boundaries”), method independence may break down.
- **Threshold effects:** For α slightly below α_{crit} , small perturbations can shift the system between method-dependent and method-independent regimes. This suggests using $\alpha \gg \alpha_{\text{crit}}$ to ensure robustness.

E.6 Comparison to Other Fairness Frameworks

Our fairness properties differ from traditional redistricting fairness criteria:

Fairness Criterion	Traditional	Edge-Weighting
Partisan fairness	Varies by method	Same across methods
VRA compliance	Varies by method	Same across methods
Compactness	Varies by method	Same across methods
Algorithmic determinism	No	Yes
Gaming resistance	No	Yes
Verifiability	Difficult	Easy

Table 7: Fairness properties under traditional vs. edge-weighted redistricting. Edge-weighting induces method-independent outcomes, enabling new fairness guarantees.

E.7 Implications for Redistricting Reform

These fairness guarantees suggest concrete policy recommendations:

1. **Standardize edge-weighting:** Redistricting commissions should adopt standardized edge weight formulas (e.g., $\alpha = 2k/m_{\text{state}}$) rather than allowing discretionary method choice.
2. **Require algorithmic transparency:** Published redistricting plans should include edge weights and partition algorithm specifications, enabling independent verification.
3. **Shift burden of proof:** Courts should presume algorithmic validity if plans match independently-verified solutions, placing burden on plaintiffs to show deviations are unjustified.
4. **Enable competition:** Multiple independent parties can produce maps using different algorithms and verify convergence, reducing reliance on single trusted map-drawers.

These recommendations align with broader calls for algorithmic accountability in government decision-making ???.

E.8 Open Questions

Several fairness-related questions remain:

- **Multi-objective fairness:** If we add constraints beyond VRA (e.g., community of interest preservation), does method independence persist?
- **Dynamic fairness:** How do fairness properties evolve across census cycles as demographic distributions shift?
- **Perceptual fairness:** Do stakeholders perceive algorithmically-determined districts as more fair than human-drawn districts, even if outcomes are identical?
- **Differential fairness:** Do fairness guarantees apply equally to all demographic groups, or do some groups benefit more from algorithmic determinism than others?

We leave these questions for future work, noting that our framework provides a foundation for analyzing fairness in algorithmic redistricting beyond traditional metrics.

E.9 Summary

Edge-weighting with $\alpha \geq \alpha_{\text{crit}}$ induces three key fairness properties: (1) algorithmic determinism (tract assignment independent of method choice), (2) gaming resistance (strategic method selection confers no advantage), and (3) transparency-fairness equivalence (simple methods achieve same outcomes as complex methods).

These properties enable verification, auditability, and burden-shifting in redistricting litigation. They suggest policy reforms including standardized edge-weighting, algorithmic transparency requirements, and independent verification protocols.

F Experimental Protocol Details

This appendix provides complete experimental methodology details for reproducibility.

F.1 Software and Hardware

METIS Configuration:

- Version: METIS 5.1.0 (March 2013 release)
- Wrapper: Python 3.11 with METIS C library via ctypes
- Algorithm: Multilevel k-way partitioning (METIS_PartGraphKway for n-way), recursive bisection (METIS_PartGraphRecursive for bisection)
- Command-line flags: `-ufactor=5` (0.5% imbalance tolerance), `-niter=10` (10 refinement iterations), `-seed=42` (reproducible random seed)
- Edge weight handling: Symmetric weights (if edge (u, v) has weight w , then edge (v, u) has weight w)

Hardware Specifications:

- CPU: AMD Ryzen 9 5950X (16 cores, 3.4 GHz base / 4.9 GHz boost)
- RAM: 64 GB DDR4-3600
- Storage: 2 TB NVMe SSD (Samsung 980 PRO)
- OS: Windows 11 Pro (Build 26200), Python 3.13

Runtime Environment:

- Single-threaded execution (METIS does not use parallelism by default)
- No other compute-intensive processes during experiments
- Total wall-clock time for all 43 runs: 47 minutes (mean: 66 seconds per run)

F.2 Data Preprocessing

Census Tract Data:

- Source: US Census Bureau 2020 Redistricting Data (P.L. 94-171)
- Downloaded: April 2021 (official release)
- Tract count by state: Alabama (1,181), Georgia (1,969), Louisiana (1,148), Mississippi (667), South Carolina (1,103)
- Total tracts: 6,068

Minority VAP Calculation:

- Voting Age Population (VAP) = Population aged 18+
- Minority VAP = Black VAP + Hispanic VAP + Asian VAP + Native American VAP + Pacific Islander VAP + Two or More Races VAP
- Non-minority (White) VAP = Total VAP - Minority VAP
- Minority percentage = Minority VAP / Total VAP

Adjacency Graph Construction:

- Method: Shared boundary detection using GeoPandas 0.12.2
- Two tracts are adjacent if their geometries share a boundary segment (not just a point)
- Adjacency is symmetric: if tract i is adjacent to tract j , then tract j is adjacent to tract i
- Average degree: Alabama: 5.8, Georgia: 5.6, Louisiana: 5.9, Mississippi: 5.7, South Carolina: 5.7
- Graph connectivity: All adjacency graphs are connected (single component)

Edge Weight Assignment:

- Minority threshold: $\tau = 0.40$ (40% minority VAP)
- Edge weight factor: $\alpha = 5.0$
- Weight formula: $w(u, v) = \alpha$ if $m_u \geq \tau$ and $m_v \geq \tau$, otherwise $w(u, v) = 1$
- Weight symmetry: $w(u, v) = w(v, u)$ for all edges (u, v)
- Minority-minority edge fraction: Alabama: 12.3%, Georgia: 18.7%, Louisiana: 15.1%, Mississippi: 14.8%, South Carolina: 11.9%

F.3 Experimental Design

Test States: Five states selected based on:

1. High minority population (Black+Hispanic VAP > 35%)
2. Active VRA Section 2 litigation history
3. Range of district counts ($k \in \{4, 6, 7, 14\}$) to test scalability
4. Geographic diversity (Deep South + Atlantic coast)

Partitioning Methods:

1. **Predetermined Recursive Bisection:** For each k , test all possible first split configurations: $(1, k-1), (2, k-2), \dots, (\lfloor k/2 \rfloor, \lceil k/2 \rceil)$. Example: $k = 7$ tests 3 configurations: $(1, 6), (2, 5), (3, 4)$.
2. **Adaptive Recursive Bisection:** Same as predetermined but evaluates all k candidate first splits and selects the one maximizing minority concentration. (In our experiments, adaptive always selected $(1, k-1)$ due to edge-weighting.)

3. **N-way Partitioning:** Single METIS call with k partitions directly (no recursive splitting).

Total Runs:

- Alabama ($k = 7$): 3 predetermined + 1 adaptive + 1 n-way = 5 runs
- Georgia ($k = 14$): 7 predetermined + 1 adaptive + 1 n-way = 9 runs
- Louisiana ($k = 6$): 3 predetermined + 1 adaptive + 1 n-way = 5 runs
- Mississippi ($k = 4$): 2 predetermined + 1 adaptive + 1 n-way = 4 runs
- South Carolina ($k = 7$): 3 predetermined + 1 adaptive + 1 n-way = 5 runs
- **Total:** 28 runs

Additional Experiments:

- Robustness test: 10 random seeds for each method times each state = 150 runs
- Tree structure experiment: All Catalan number trees for $k \in \{4, 5, 6\}$ = 65 additional runs
- **Grand Total:** 243 runs

F.4 Randomness and Reproducibility

METIS Random Seed:

- Fixed seed: `-seed=42` for all runs in main experiment
- Multiple seeds: `-seed={1,2,3,...,10}` for robustness test
- Seed affects: tie-breaking during coarsening, initial partition during refinement
- Seed does *not* affect: graph structure, edge weights, balance constraints

Single Run Policy: For the main experiment, we report results from a *single run* per method per state using `-seed=42`. This is justified because:

1. Edge-weighting with $\alpha = 5$ induces deterministic behavior
2. Robustness test shows variance across seeds is negligible ($< 0.1\%$)
3. Single-run policy matches prior graph partitioning literature ??

For the robustness test, we average over 10 seeds and report mean $\pm$ standard deviation.

F.5 Metrics Calculation

Primary Metrics:

- **Max Minority Percentage:** $\max_{i \in [k]} \frac{M_i}{N_i}$ where M_i is minority VAP in district i and N_i is total VAP in district i
- **MM District Count:** Number of districts with minority percentage $\geq 50\%$ (majority-minority districts under VRA)
- **Population Balance:** Maximum deviation from target population. Constraint: $\leq 0.5\%$ (METIS `-ufactor=5`)

Secondary Metrics:

- **Polsby-Popper Score:** $PP(D) = \frac{4\pi \cdot \text{Area}(D)}{\text{Perimeter}(D)^2}$, computed using GeoPandas `.area` and `.length` methods on dissolved tract geometries
- **Objective Value:** Weighted edge-cut = $\sum_{(u,v) \in \text{cut edges}} w(u,v)$, where cut edge means u and v are in different districts
- **Runtime:** Wall-clock time from start of METIS call to return, measured using Python `time.perf_counter()`

F.6 Statistical Analysis

Variance Calculation: For each state, compute variance across methods:

$$\text{Var}(X) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

where n is number of methods (5-9 depending on state) and x_i is max minority percentage for method i .

Significance Testing: We do not perform statistical significance tests (e.g., t-tests, ANOVA) because:

1. Sample size is small ($n \leq 9$ methods)
2. Observed variance is essentially zero ($< 0.1\%$), making significance testing unnecessary
3. Determinism is theoretical, not empirical statistical inference

Reporting Precision:

- Minority percentages: 1 decimal place (e.g., 66.2%)
- Compactness scores: 2 decimal places (e.g., 0.42)
- Runtimes: integer seconds (e.g., 83 seconds)
- Variances: 4 decimal places (e.g., 0.0001)

F.7 Code Availability

All code, data, and results are available at:

- GitHub repository: [https://github.com/\[anonymized-for-review\]](https://github.com/[anonymized-for-review])
- Includes: Python scripts, METIS wrapper, tract data processing, visualization code
- License: MIT (open source)

- Documentation: README with step-by-step reproduction instructions

Reproduction Steps:

1. Download census tract shapefiles and redistricting data
2. Run `build_adjacency.py` to construct graphs
3. Run `compute_edge_weights.py` with $\alpha = 5$, $\tau = 0.40$
4. Run `run_partitioning.py` for each method and state
5. Run `analyze_results.py` to compute metrics
6. Run `create_visualizations.py` to generate figures

Expected runtime: < 2 hours on modern hardware.

References

Thornburg v. Gingles, 478 U.S. 30. 1986. U.S. Supreme Court.